Jessica Imlau Dagostini

jessicadagostini@gmail.com jessica.dagostini@ucsc.edu
Personal Page Linkedin GitHub

Areas of Interest

High-Performance Computing | Optimization | Scientific Computing | Load Balancing

Education

Sep,2022 – Present

Ph.D. Candidate in Computer Science
University of California Santa Cruz (UCSC), Santa Cruz, United States
Expected Graduation Date: June, 2028
Advisor: Prof. Dr. Abel Souza

Mar,2020 – Mar,2022

M.Sc. in Computer Science (GPA: A)
Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil
"Performance Improvements Applied in an Electromagnetic Inversion Application Focused on Homogeneous and Heterogeneous Computational Environments"
Advisor: Prof. Dr. Lucas Mello Schnorr

Feb,2015 – Dec,2019

B.Sc. in Computer Science (Summa Cum Laude – 9.39 / 10)
Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil
"Evaluating the Container's Applicability in a Distributed System of Code Judgement"
(Avaliação da Aplicabilidade de Containers em um Sistema Distribuído de Julgamento de Códigos.)
Advisor: Prof. Ms. Neilor Avelino Tonin
Co-advisor: Dr. Jean Luca Bez

Honors & Fellowships

Aug, 2024 **2024 ACM SIGHPC Computational and Data Science Fellowship**
Highly competitive fellowship sponsored by ACM SIGHPC focused on underrepresented groups working with High Performance Computing. 2024 cohort composed only by two fellows.

Apr, 2024 **CRA-WP Grad Cohort for Women**
Selected student to participate in the annual Grad Cohort for Women on Minneapolis, MN.

2023 **2º Place - ISC Virtual Student Cluster Competition**
Team Not-So-Slow-Slugs - UCSC. Awarded by ISC High Performance 2023 Conference.

Apr,2023 – Jun,2023 **UCSC Dean's Fellowship**
Academic tuition coverage given to a small number of outstanding admitted PhD students at UCSC.

2021 **Students@SC'21 Best Lightning Talk**
Awarded by Students@SC Program from Supercomputing'21 Conference

Mar,2020 – Mar,2022 **Master Scholarship**
Financial scholarship provided by Petroleo Brasileiro, to work in the research project "Otimização Computacional de Códigos de Inversão de Dados" (Computational Optimization of Data Inversion Codes)

2020 **URI Erechim Best Student Award** Summa Cum Laude (9.39 / 10.0)
Awarded by Universidade Regional Integrada do Alto Uruguai e das Missões (URI)

2020 **SBC Highlight Student Award** Summa Cum Laude (9.39 / 10.0)
Awarded by Brazilian Computer Society (SBC)

2016 **2º Place - Tecnomate Contest 2016, Categoria "Libres"**
Awarded by Departamento de Ingeniería en Sistemas de Información, UTN Santa Fé (Argentina).

Fev,2015 – Dez,2019 **University for All Program (ProUni)**
Financial and merit-based full scholarship provided by Brazilian Federal Government to cover university tuition (selected as 1st place)

2010 **Honorable Mention on 6th Brazilian Math Olympiads of Public Schools (OBMEP)**
Awarded by Conselho Nacional de Pesquisa (CNPq), Brasil.

Professional Experience

Sept,2022 – Ongoing **Graduate Student Researcher**

University of California, Santa Cruz, Santa Cruz, United States
Advisor: Prof. Dr. Abel Souza

- Developing an intent-driven system for scientific computing, to aid in the configuration-decision when running scientific applications. Intents in this context are high-level computational goals when running a scientific application (such as run in less than X hours, use less than Y node hours, etc).
- Defining the intent-driven specification to allow different application to benefit from the system.

Co-advisor: Prof. Dr. Scott Beamer
Co-advisor: Prof. Dr. Tyler Sorensen

- Development of a proxy application based on this pangenome mapping tool to allow the exploration of software improvements
- Understanding the impact of load balancing algorithms on power-efficiency usage on distributed systems

Jan, 2023 – Ongoing **Teaching Assistant**

University of California, Santa Cruz, Santa Cruz, United States
Courses:

- CSE113 - Parallel and Concurrent Programming (Winter 2023, Winter 2024, Fall 2024, and Winter 2026)
- CSE120 - Computer Architecture

Jun, 2026 – Aug, 2026 **Science Internship Project Mentor**

University of California, Santa Cruz, Santa Cruz, United States
Project: AI Intent ”Translator”: Using AI to Turn Natural Language into Supercomputer Job Settings

- Mentored a group of 3 high school students in a 10-week program
- Students searched about LLM models, how to create agents and how to evaluate their behavior
- The goal was to build an agent that translate natural language high-level computational goals into structured queries to search for configurations that achieved the specifies goal

June,2025 – Sept, 2025 **Graduate Student Research Assistant Intern**

National Energy Research Scientific Computing Center (NERSC)
Lawrence Berkeley National Lab
Mentor: Dr. Kevin Gott

- Development of a toolbox to extract load balancing metrics from real simulation from WarpX, a particle-bsaed application that utillizes AMReX, allowing the easy testing of different load balancing strategies
- Implementation of support of memory capacity as a constraint in the load balancing algorithms (with modifications in both AMReX and WarpX source-code)

Sept,2023 – Dec,2023 **PhD Intern – High Performance Computing**

Pacific Northwest National Laboratory, Richmond, United States
Advisor: Dr. Joseph B Manzano

- Performance characterization of a DNA mapping application that uses pangenomes as references and a graph structure called GBWT (Graph Burrows-Wheeler Transform) to index DNA readings
- Started working towards developing a proxy application based on this pangenome mapping tool

Jan,2021 – Jul, 2022 **Principal System Architect**

beecrowd, São Paulo, Brazil

- Responsible for the development and maintenance of the programming language support and setup of the beecrowd judging system;
- Organize and supervise team’s activities.

Mar,2020 – Mar, 2022 **Graduate Student Researcher**

Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil
"Computational Optimization of Data Inversion Codes"
(Otimização Computacional de Códigos de Inversão de Dados)
Advisor: Prof. Dr. Lucas Mello Schnorr

- This project investigated and implemented computational techniques that improved the execution and quality of results of Data Inversion codes;
- Evaluated, proposed and applied algorithms to improve the load balancing and scalability of a data inversion application of survey oil and gas in homogeneous and heterogeneous systems.

May,2019 – Sep,2019 **Adjunct Faculty**

National Service for Commercial Education - RS, SENAC/RS, Erechim, Brasil.

- Taught robotics to K-12 students in a project complementary to school;
- Teacher in two components of the Fundamentals of Informatics course (Compact Internet and Compact PowerPoint).

Jan,2019 – Mar,2019 **Intern Researcher (Summer Program)**

Brazilian Center for Research in Energy and Materials (CNPEM), Campinas, Brazil
Brazilian Synchrotron Light Laboratory (LNLS), Brasil
"GPU Code Implementation to Phase Recovery on Fraunhofer Region"
(Implementação em GPU para a Recuperação da Fase na Região de Fraunhofer)
Advisor: Dr. Eduardo Xavier Miqueles

- Summer internship project consisted of the development of a specific tool for the restoration of projective images obtained in the Fraunhofer region;
- Main developer of an application in CUDA to be incorporated into the pipeline of the automatic processing of the detectors from the Sirius synchrotron light source;
- Discovered some bugs at cuFFT library that were reported to NVIDIA developers.

Aug,2018 – Jul,2019 **Undergraduate Student Extensionist**

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil
"Insertion of High School Students from Erechim on Brazilian Olympiads in Informatics"
(A Inserção de Alunos do Ensino Médio de Erechim na Olimpíada Brasileira de Informática)
Advisor: Prof. Ms. Neilor Avelino Tonin

- Project to teach algorithms and programming languages to High School students
- Taught programming to 45 students that enrolled in the Brazilian Olympiads in Informatics (OBI)

Aug,2015 – Jul,2018 **Undergraduate Research Assistant**

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil

- *"A Study About Distributed Systems to be Applied on URI Online Judge Portal"*
  (Um Estudo Sobre Sistemas Distribuídos para Aplicação no Portal URI Online Judge)
  Advisor: Prof. Ms. Neilor Avelino Tonin
  - Distributed the front-end servers from URI Online Judge;
  - Studied and applied tests to verify which system distribution will be the best option for the environment
- *"Developing a Customized Online Forum to URI Online Judge"*
  (Criação de um Fórum Customizado para o Portal URI Online Judge)
  Advisor: Prof. Ms. Neilor Avelino Tonin
  - Rebuilt the online forum from the URI Online Judge (now beecrowd) platform, applying some concepts related to Human-Computer Interaction, developed using CakePHP framework and PostgreSQL database;
  - The new version was deployed in April, 2017, now being used by 300k users, with more than 8800 posts (*https://www.beecrowd.com.br/judge/en/forum*).

Apr,2012 – Aug,2014 **Junior Student Research**

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil
Escola Estadual Normal José Bonifácio (High School), Brazil
"Discussion and Deepening of Concepts Regarding the Study of Mathematical Functions on High School"
(Discussão e Aprofundamento de Conceitos a Respeito do Estudo de Funções do Ensino Médio)
Advisor: Prof. Ms. Clemerson Alberi Pedroso

- Investigated, discuss, and deepen mathematical concepts related to the study of functions and contribute to the approximation of scientific-theoretical knowledge didactic-operational knowledge in Exact Sciences;
- Developed a diversity of studies from applied mathematics and created a collection of problems modeled from real-world problems.

Publications

Full Papers Published in Conferences

- **Dagostini, J.**; Manzano, J., Sorensen, T., Beamer, S. **miniGiraffe: a Pangenome Proxy App**. In: *2025 IEEE International Symposium on Workload Characterization (IISWC)*, Irvine, CA, USA, 2025, pp. 339-352, https://doi.org/10.1109/IISWC66894.2025.00036.
- **Dagostini, J.**; H. C. P. da Silva, V. G. Pinto, R. M. Velho, E. S. L. Gastal and L. M. Schnorr. (2021) **Improving Workload Balance of a Marine CSEM Inversion Application.** In: *2021 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, 2021, pp. 704-713, https://doi.org/10.1109/IPDPSW52791.2021.00107.
- **Dagostini, J.**; Lima, M. V. de M.; Bez, J. L.; Tonin, N. A. (2018) **URI Online Judge Blocks: Construindo Soluções em uma Plataforma Online de Programação.** In: *XXIX Simpósio Brasileiro de Informática na Educação (Brazilian Symposium on Computers in Education)*, Fortaleza, 2018. v. 1. p. 168-177. http://dx.doi.org/10.5753/cbie.sbie.2018.168

Short Papers and Posters Published in Conferences

- Yellapragada, S., **Dagostini, J.**, Gott, K., Hartman-Baker, R. (2025) **Heterogeneity-Aware Task Allocation for Modern HPC Systems.** In *SC25: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, St. Louis, MO, USA. **Best Research Poster Finalist.**
- **Dagostini, J.**, Yellapragada, S., Gott, K., Hartman-Baker, R. (2025) **1** In *SC25: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, St. Louis, MO, USA.
- **Dagostini, J.**; Pinto, V., & Schnorr, L. (2021). **Aplicação de Método Guloso para Balanceamento de Carga em uma Aplicação de Exploração Eletromagnética.** In: *Anais da XXI Escola Regional de Alto Desempenho da Região Sul*, (pp. 87-88). Porto Alegre: SBC. https://doi.org/10.5753/eradrs.2021.14782
- **Dagostini, J.**; Lima, M. V. de M.; Bucior, L; Bez, J. L.; Tonin, N. A. (2017) **Incentivando a Aprendizagem de Algoritmos Através do URI Online Judge Forum 2.0** In: *XXVIII Simpósio Brasileiro de Informática na Educação (Brazilian Symposium on Computers in Education)*, Recife, 2017. v. 1. p. 1781. http://dx.doi.org/10.5753/cbie.sbie.2017.1781

Presentations and Invited Talks

Invited Talks

1. **Dagostini, J.**; Beamer, S., & Sorensen, T. , & Manzano, J., & Marques, A. (2024). **Developing a Proxy Application Based on a Parallel Pangenome Mapping Tool.** In: *The 1st Workshop on Emerging Computer Systems Challenges and Applications in Biomedicine (BioSys).* San Diego, CA, USA.
2. **Dagostini, J.**; Solórzano, A. L.; Girelli, V.; Carmin, M. (2022) **Special Sessions - Women in High Performance Computing**. *(WHPC Affiliate Brasil - Região Sul. XXII Escola Regional de Alto Desempenho, Parana, Brasil*

3. **Dagostini, J.**; Gayler, R.; Beste E. de;F ratkin, M.; Merritt, C.; Gonsiorowski, E.; Belcher, K.; Chen, D. (2021) **Addressing Challenges in HPC Careers**. *Students@SC Student Programming, Supercomputing 21 Conference, St. Louis, United States*

4. **Dagostini, J.**; Baptista, V.; Zhan, Y.; Leiva, F. H.; Gyawali, S.; Johnston, B.; Filinger, W. (2021) **Presentation and Communication Skills**. *Students@SC Student Programming, Supercomputing 21 Conference, St. Louis, USA*

5. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Por que ainda precisamos falar sobre mulheres em HPC?**. *English title: Why do we still have to talk about women in HPC? (WHPC Affiliate Brasil - Região Sul) WSCAD Conference, Minas Gerais, Brasil*

6. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Latin American Women on HPC: WHPC Affiliate Brasil Região Sul**. *Latin America High Performance Computing Conference (CARLA 2021), Guadalajara, Mexico*

7. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Special Session - Women in High Performance Computing**. *(WHPC Affiliate Brasil - Região Sul) XXI Escola Regional de Alto Desempenho, Santa Catarina, Brasil*

8. **Dagostini, J.**; Lima, M. V. de M.; Tonin, N. A. (2017) **Enseñar a programar jugando: URI Judge Online**. *WTIC, Resistencia, Argentina*

9. **Dagostini, J.**; Bez, J.L.; Tonin, N. A. (2016) **Portal URI Online Judge - Internacionalização. 2016.** *Tecnomate, Santa Fé, Argentina.*

Tutorials and Short Term Courses

May,2022 **Orquestrando Containers com Docker e LXC**
Cursos de Outono 2022
Short Term Course

Apr,2021 **Are you Root? Experimentos Reprodutível em Espaço de Usuário**
XXI Escola Regional de Alto Desempenho da Região Sul
Short Term Course

Oct,2020 **Introdução a Programação Paralela com OpenMP**
I Semana Acadêmica Integrada dos Cursos de Ciência da Computação da URI
Short Term Course

Community Services

Committee

2022 – 2025 **Women in HPC Workshop**
2024, 2025, and 2026 - Workshop Co-Chair
2022 and 2023 - Submissions Co-Chair

2026 **ACM/IEEE SC'26 Supercomputing Conference, Chicago, IL (USA)**
Artifact Description / Article Evaluation (AD/AE) Appendix Reviewer

2023 **ACM/IEEE SC'23 Supercomputing Conference, Denver, CO (USA)**
Infrastructure - Wayfinding Co-Chair
Artifact Description / Article Evaluation (AD/AE) Appendix Reviewer
Lead Student Volunteers Reviewer

2023 **International Conference on Parallel Computing**
Reproducibility Committee

2022 **29th IEEE Hot Interconnects Symposium**
Web Chair

2019 **X Simpósio de Tecnologia da Informação da Região Norte e Noroeste do Rio Grande do Sul XXIII Semana Acadêmica do Curso de Ciência da Computação da URI Erechim, Brasil.**
Member of the Organizer Team

2019 **IX Simpósio de Tecnologia da Informação da Região Norte e Noroeste do Rio Grande do Sul XXII Semana Acadêmica do Curso de Ciência da Computação da URI Erechim, Brasil.**
Member of the Organizer Team

2019 **III Engenharia em 24 Horas: o desafio. (URI Erechim, Brasil)**
Member of the Organizer Team

External Reviwer

2022 – Present **Parallel Computing Jounal**

Volunteering

Nov,2020 – Nov,2025 **ACM/IEEE SC Conference Series**

- *SC'25* Selected Lead Student Volunteer (Students@SC HQ)
- *SC'24* Selected Lead Student Volunteer (Students@SC Networking)
- *SC'22* Selected SCALE/Lead Student Volunteer (Wayfinding Committee)
- *SC'21* Selected SCALE/Lead Student Volunteer (Communication Committee)
- *SC'20* Selected Regular Student Volunteer

Jan,2021 – Dec, 2022 **Women in HPC Affiliate Brasil - Região Sul**
Co-Founder of the Women in HPC Affiliate Brasil - Região Sul.
First representation of WHPC Global in Latin America.

Jan,2022 – Dec,2022 **BRASA PRÉ Pós-Graduação Mentor**
Mentor of a Brazilian student for the application process for graduate schools in USA.

Apr,2022 **Volunteer Staff**
2022 International Collegiate Programming Contest (ICPC)
South America/Brazil Finals
Principal Activities:

- Help with general organization and coordination in the whole final contest event.

May,2015 – Aug,2017 **Volunteer Member**
Rotaract Club Erechim Paiol Grande - Rotaract Club recognized by the Rotary International

Skills

Environments

- Working proficiency: Linux/Unix | nano | bash | git | Docker and LXC containers | LaTeX;
- Basic proficiency: spack | slurm | vi.

Programming Languages

- Working proficiency: C | C++ | Python | R | PHP | HTML/CSS;
- Basic proficiency: Java | JavaScript | Ruby.

APIs and Frameworks

- Working proficiency: MPI | OpemMP | CakePHP | Score-P;
- Basic proficiency: CUDA | POSIX Threads | OpenACC.

General Skills

- Abilities with leadership and teamwork;
- Experience with teaching algorithms to new programmers;
- Knowledge of teaching didactics (High School with disciplines for teaching degree).